

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	1 October 2026
Team ID	6ab7f31f8c7c66e70672939b
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: EduGenie – AI-powered personalized learning assistant for students

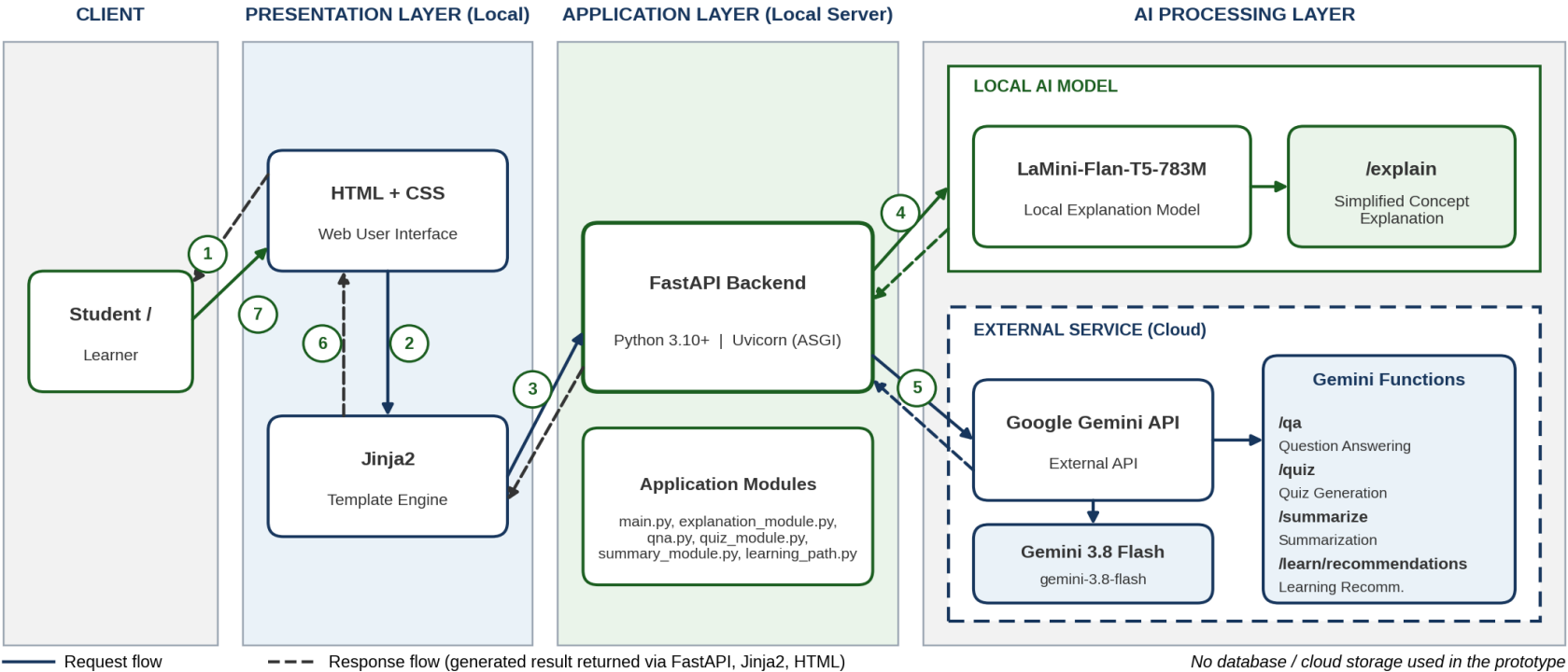


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web UI through which students interact with EduGenie and access the explanation, Q&A, quiz, summarization and learning recommendation features	HTML, CSS
2.	Template Engine	Renders dynamic HTML pages from the FastAPI backend and integrates generated results into the web interface	Jinja2
3.	Application Logic-1 (Request Handling)	Handles HTTP requests, routing, input processing and coordination of application modules (main.py)	FastAPI, Python 3.10+
4.	Application Logic-2 (ASGI Server)	Runs and serves the FastAPI application	Uvicorn
5.	Application Logic-3 (Explanation)	Generates simplified explanations of difficult academic concepts – /explain (explanation_module.py)	LaMini-Flan-T5-783M
6.	Application Logic-4 (Q&A)	Answers student questions and clarifies academic doubts – /qa (qna.py)	Google Gemini API, Gemini 3.8 Flash
7.	Application Logic-5 (Quiz)	Generates topic-based quizzes and practice questions – /quiz (quiz_module.py)	Google Gemini API, Gemini 3.8 Flash
8.	Application Logic-6 (Summarization)	Produces concise summaries of lengthy learning content – /summarize (summary_module.py)	Google Gemini API, Gemini 3.8 Flash
9.	Application Logic-7 (Learning Recommendations)	Provides structured learning path and topic recommendations based on the learning context – /learn/recommendations (learning_path.py)	Google Gemini API, Gemini 3.8 Flash
10.	Database	No persistent database is currently required for the implemented prototype	Not Used / N/A
11.	Cloud Database	No cloud database service is used	Not Used / N/A
12.	File Storage	No dedicated cloud file storage is required; application files and resources are kept locally	Local Filesystem / Application Resources
13.	External API	Google Gemini API provides cloud-based generative AI capabilities for Q&A, quiz generation, summarization and learning recommendations	Google Gemini API
14.	Machine Learning Model-1	Local model that generates simplified explanations of complex academic concepts (/explain)	LaMini-Flan-T5-783M
15.	Machine Learning Model-2	General-purpose generative AI model for Q&A, quiz generation, summarization and learning recommendations; accessed through the external Google Gemini API (model ID: gemini-3.8-flash)	Google Gemini 3.8 Flash
16.	Infrastructure (Server / Cloud)	Application deployment on Local System. Local Server Configuration: FastAPI application served by Uvicorn on a local server / development environment. Cloud Server Configuration: Not currently specified / Not required for the prototype	Local Server / Development Environment

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	EduGenie uses open-source Python web / application technologies for the backend and templating layer	FastAPI, Jinja2, Uvicorn, Python ecosystem
2.	Security Implementations	Implemented / design-level: Gemini API key is kept in environment variables and not hard-coded in source files; request validation is applied at the backend. Recommended for deployment (not currently implemented): HTTPS, controlled API access. Encryption, IAM, firewalls, OAuth and advanced authentication are not implemented	Environment Variables, FastAPI Request Validation, HTTPS (when deployed)
3.	Scalable Architecture	Current prototype: logical 3-tier design – Presentation Layer, Application Layer and AI Service Layer. Potential production scalability: the FastAPI backend can be run behind a suitable production server configuration and scaled horizontally when required. Horizontal scaling, load balancing, Kubernetes and micro-services are not currently implemented	FastAPI, Uvicorn, Google Gemini API
4.	Availability	Current prototype: availability depends on the local server and on the availability of the external Google Gemini API. Production deployment: availability can be improved using multiple application instances, health checks and load balancing (recommended, not currently implemented)	FastAPI, Uvicorn, Google Gemini API
5.	Performance	Design considerations: lightweight HTML / CSS frontend, FastAPI asynchronous request handling, efficient API request processing, external Gemini API latency, local LaMini model inference time, response size and concurrent request handling. No benchmark values have been measured	FastAPI, Uvicorn, LaMini-Flan-T5-783M, Google Gemini 3.8 Flash

Table-3: AI Model Responsibility:

AI Model	Responsibility	Related Functionality
LaMini-Flan-T5-783M (Local)	Simplified Concept Explanation	/explain
Google Gemini 3.8 Flash (External API)	Question Answering	/qa
Google Gemini 3.8 Flash (External API)	Quiz Generation	/quiz
Google Gemini 3.8 Flash (External API)	Summarization	/summarize
Google Gemini 3.8 Flash (External API)	Learning Recommendations / Structured Learning Path	/learn/recommendations

Technology Stack Summary:

Layer / Item	Technology
Frontend	HTML, CSS
Template Engine	Jinja2
Backend	FastAPI
Programming Language	Python 3.10+
ASGI Server	Uvicorn
Local AI Model	LaMini-Flan-T5-783M
Cloud Generative AI	Google Gemini API
Gemini Model	Google Gemini 3.8 Flash (gemini-3.8-flash)